

TOP ALGORITHMS IN HEALTHCARE

Naïve Bayes - Based on the Bayes theorem used to determine probability of an event based on prior knowledge of conditions, this is one of the most efficient machine learning algorithms in use currently. All clinicians, both medical doctors and clinical psychologists, have been taught Bayes' rule at university. It's abundantly used in the healthcare industry for designing systems that fundamentally help improve the quality of data leading up to disease prediction.

Support Vector Machine (SVM)

- SVM is a critical supervised machine learning model that's capable of handling various data sets to resolve multiple problems, especially in a healthcare setting. According to experts, classification of medical data is the primary way SVM can do medical data analytics – and this very fundamental classification of medical data is vital for clinical coding and transforming the medical data into standardised statistical code. Researchers around the world have verified SVM's ability to better detect prevailing health conditions such as cancer, blood pressure, and diabetes – simply through medical data analytics.



Futuristic doctor's face

Random Forest - Random forest is a commonly-used machine learning algorithm which combines the output from multiple sources to reach a single result. Its ease of use and flexibility have fueled its adoption, as it handles both classification and regression problems, according to IBM. The random forest algorithm has applications within computational biology, allowing doctors to make estimates around drug responses to specific medications, based on gene classification and biomarkers.

Logistic Regression

- According to IBM, this type of statistical model is often used for classification and predictive analytics. Logistic regression estimates the probability of an event occurring based on a given dataset of independent variables. Specifically in the medical field, this analytics approach can be used to

predict the likelihood of disease or illness for a given population pretty effectively. Everyone from healthcare organisations to general physicians in private practice can set up preventative care for individuals that show higher propensity for specific illnesses, using logistic regression models.

Scientist, Dr Junaid Bajwa believes we are living in a moment where technology, data and consumer demand is reshaping the healthcare market and redefining how value is defined and captured. This convergence of technology and medicine today, in delivering increasing levels of automation and soon increasing levels of artificial intelligence (AI), will lead to an autonomous future for healthcare.

"Medical knowledge doubles every 73 days," according to Dr Junaid Bajwa, when he spoke to us at the Global AI Summit, Riyadh. Bajwa believes OpenAI's GPT-3 could hold the answer to solving various challenges in healthcare, offering more specific, personalised, and result-backed healthcare solutions, treatments, and consultations to people across the globe.

With the World Health Organisation predicting a shortfall of 14-million

healthcare workers by 2030, Dr Bajwa was hopeful of the role that technology can play in fixing the gap. "Healthcare is exorbitantly expensive with 5% of the population taking anywhere between 25-60% of those resources from the most vulnerable in society. We need to move to a healthcare system that is not just about diagnosis and treatment, but about recognising and prevention,

and ultimately towards personalisation and personal healthcare," he said during a medical summit in the Middle East last year.

Dr Bajwa obviously feels AI can play a huge role in improving healthcare all around the world, but he's acutely aware of how AI requires access to data, access to domain expertise and access to massive computing power to work.

"Today's story is one of automation of processes, aggregation of data, moving to intelligent analysis and AI, and then repeating that cycle. If we get this right, it really has the potential to reduce costs and support clinicians by unmasking occult disease types, generalising new associations and perhaps even generating new novel hypotheses and new mechanisms with which we diagnose disease in ways that we could have only imagined in the past," he summed up his thoughts. **d**



DR JUNAID BAJWA